ONE MODEL. SIMULATE. FIT. OPTIMIZE.

Hands-On with Rumoca

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Purdue University · School of Aeronautics and Astronautics

Georgia Tech · 10:00–13:00

Any modeling domain welcome · no aerospace background needed

MODEL
Declare the physics

FIT
Learn from data

OPTIMIZE
Plan the motion

CREATE CORE CODESPACE

Scan to create the core Codespace · repository link above

Repository ↗ **Workshop guide ↗** **Create Codespace ↗**

Run the core. Explore your own question.

YOUR QUESTION. YOUR EXPERIMENT.

Make something you care about.

Experiment in a model or simulator.

Core: edit a model, fit data, optimize a trajectory

One change in an existing model or simulator is enough

THE PRIZE

FREE

CogniPilot stocking cap

Short demos
Audience hand vote

Enter by 12:40 · up to 5 demos · CHALLENGE.md · hosting optional

Challenge rules ↗ **Example prompts ↗**

Why Rumoca? One model, many questions.

A unified, declarative foundation for cyberphysical systems.

Lie group theory

Represent rotations and motion with their geometric structure.

Reachable sets

Bound possible behavior of cyberphysical systems under uncertainty.

Formal software verification

Connect model equations to explicit proof obligations.

System identification + digital twins

Fit physical parameters; compare simulation with experiments.

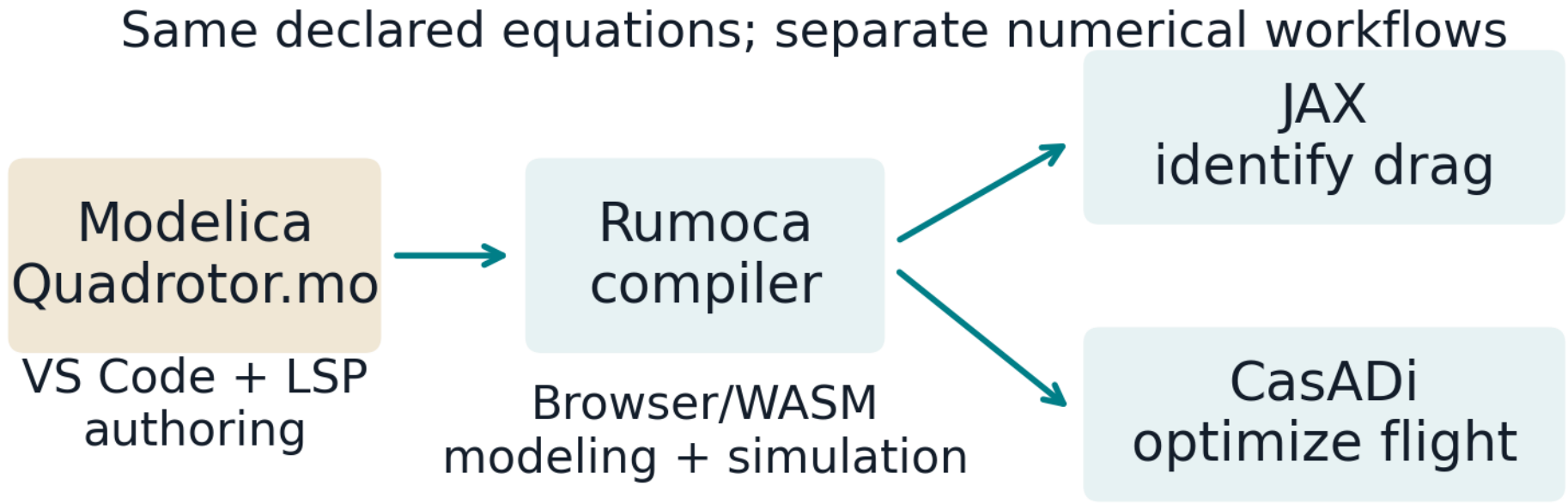
Research directions: fast Monte Carlo / RL · native AD · heterogeneous compute

Today: one Modelica model → simulation → identification → optimization.

Rumoca compiler ↗ | Model library ↗

Declare once. Ask different questions.

Quadrotor.mo → JAX + CasADi · the same declared equations

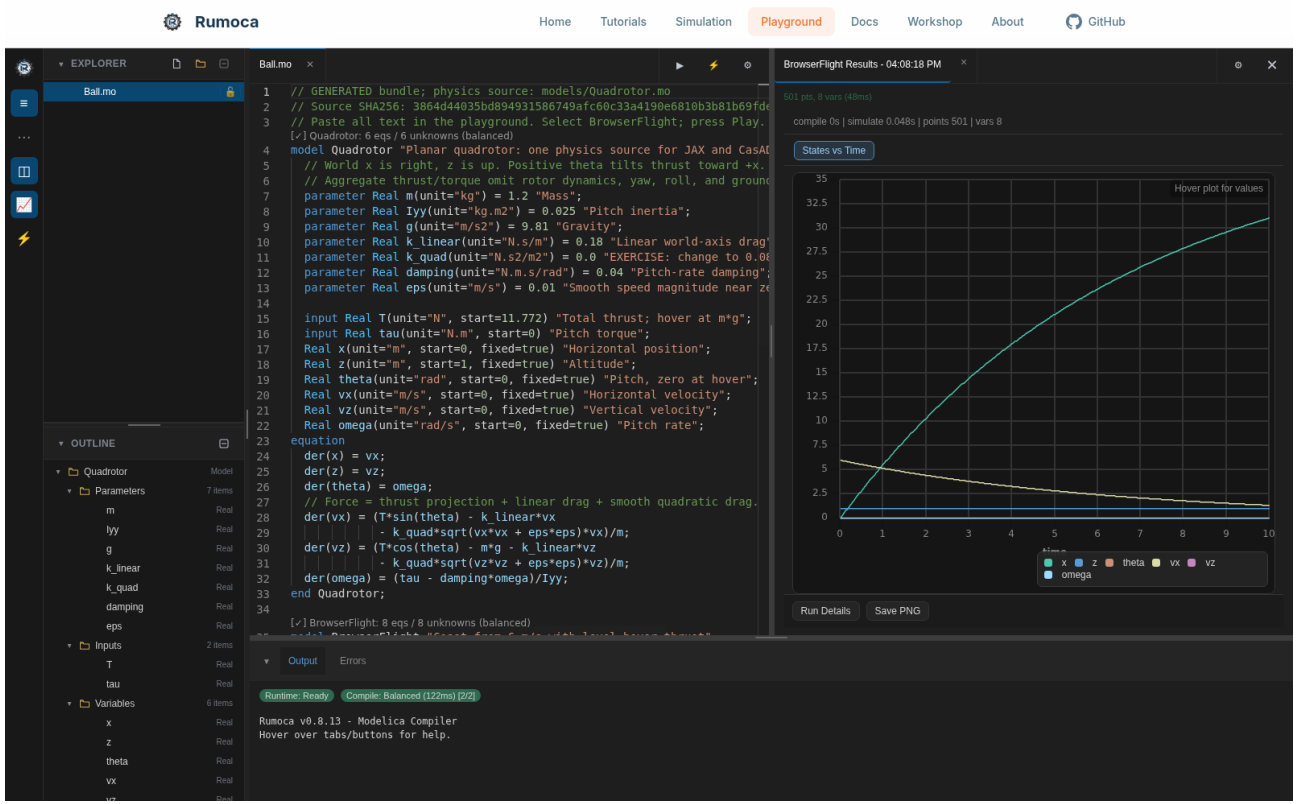


Modelica declares physics · Rumoca compiles · backends compute.

Everyone gets a browser plot

TRY THIS

1. Open the playground
2. Paste the complete browser bundle
3. Gear → BrowserFlight → Play



You should see BrowserFlight Results and a plot.

Open playground ↗

Browser model ↗

Choose a full-workshop route

01 Browser: modeling and predictions; no account

02 Codespaces: Python + VS Code; login and allowance

03 Local uv: Python + VS Code; pinned CPU packages

04 Pair or screenshare: keep working if setup fails

Switch route after ten minutes of setup trouble.

Local VS Code setup ↗ | Create Codespace ↗ | Recovery help ↗

Configure the experiment, then run it

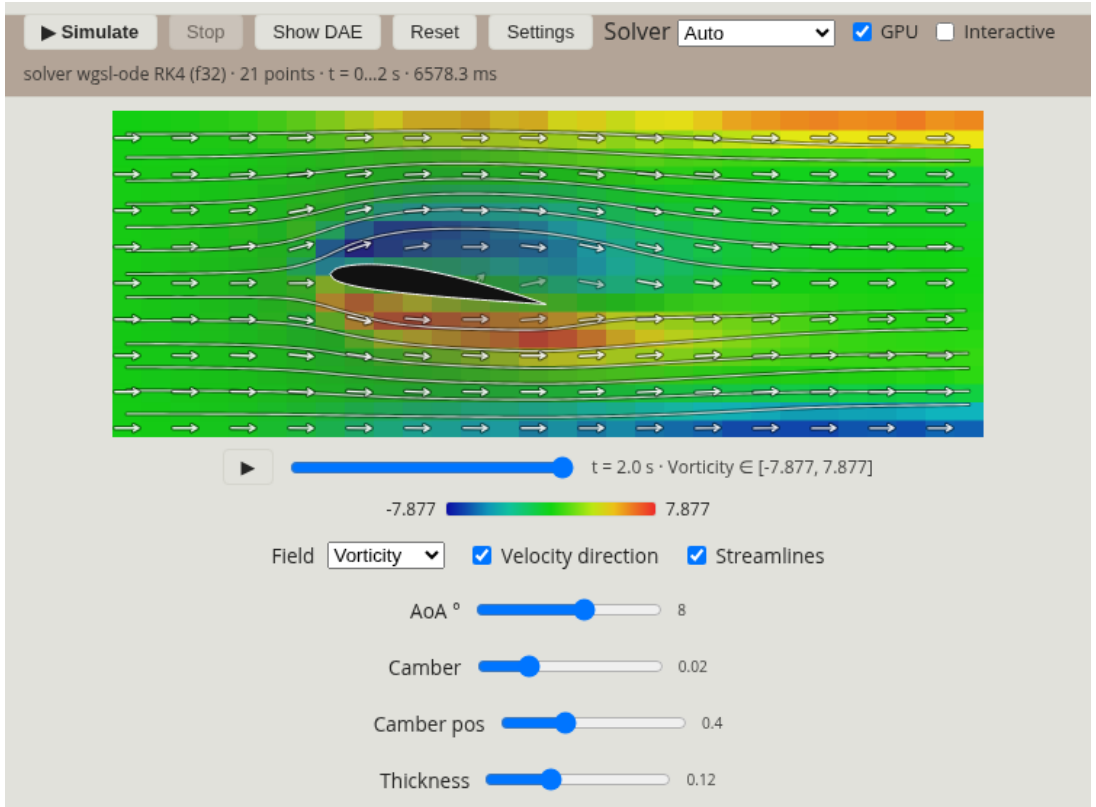
1. Open BrowserFlight.mo → click the lightning bolt
2. Save scenario TOML → model, source roots, solver, end time
3. Run the scenario → inspect vx and x

Modelica owns physics · TOML owns the run configuration.

The user guide is a live modeling lab

TRY THIS

1. Edit StopTime: 30 → 2 s
2. Simulate; select the final frame
3. Switch vorticity / pressure



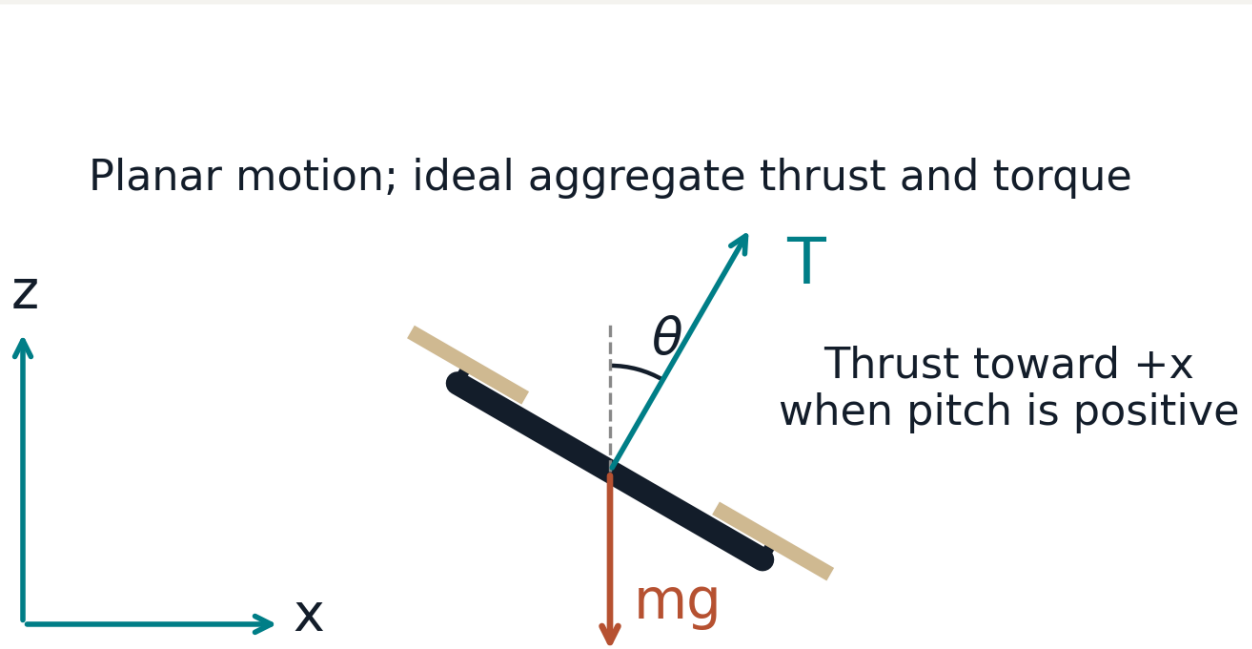
User-guide link and recovery: WORKSHOP.md · no install needed

Open NACA example ↗ **Explore user guide ↗**

A planar quadrotor we can inspect

READ THE MODEL

1. World x right; z up
2. Positive pitch tilts thrust toward +x
3. Hover: T = mg = 11.772 N



Find velocities, pitch rate, thrust and torque in the model.

Open vehicle model ↗

Workshop guide ↗

Your first engineering change: drag

Change k_quad = 0.0 to 0.08, save, then simulate.

$$D(v) = -k_{\text{linear}} v - k_{\text{quad}} \sqrt{v^2 + \varepsilon^2}\, v$$

Linear: N s/m Quadratic: N s²/m² ε = 0.01 m/s

Nonnegative coefficients ⇒ drag opposes motion

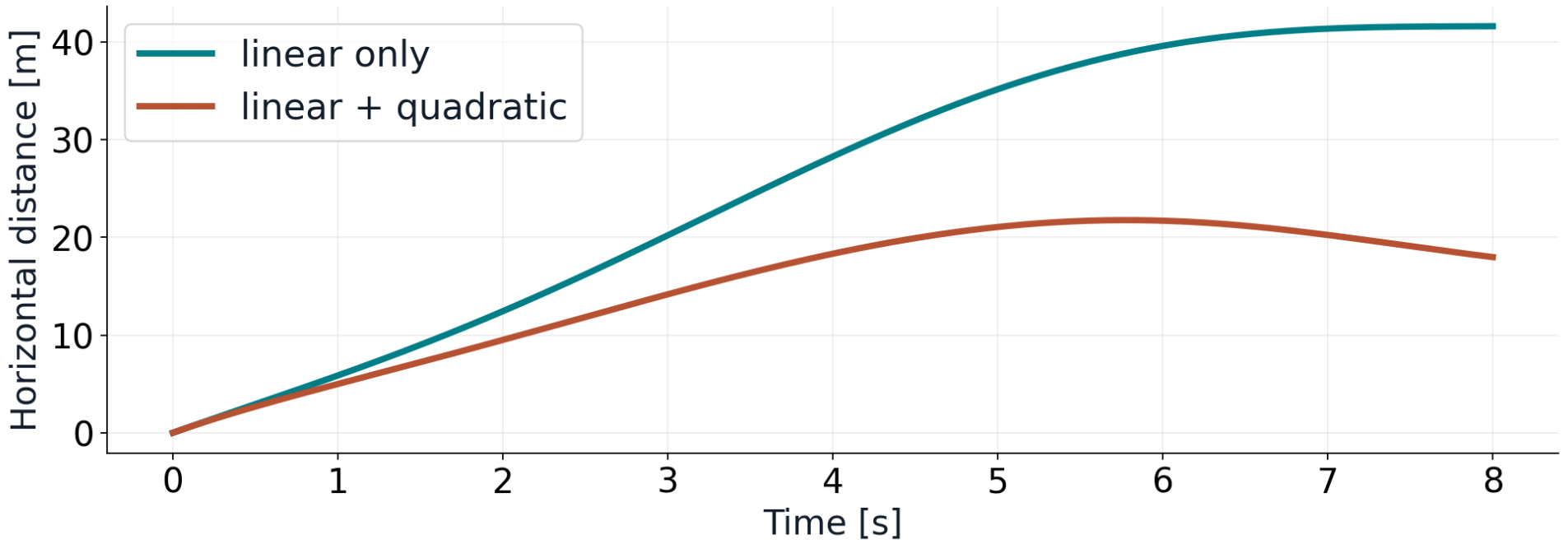
```
$ uv run --frozen python examples/simulate.py
```

What changes at high speed? What about negative velocity?

Workshop guide ↗ | Open playground ↗

More drag, less distance

Reference at 8 s: 41.60 m → 17.97 m



Compare your current default with the two reference cases.

Workshop guide ↗

Back at

11:10

10 MINUTES

Stretch. Ask a question. Keep your editor open.

KEEP YOUR PLACE

Keep your editor and terminal open.

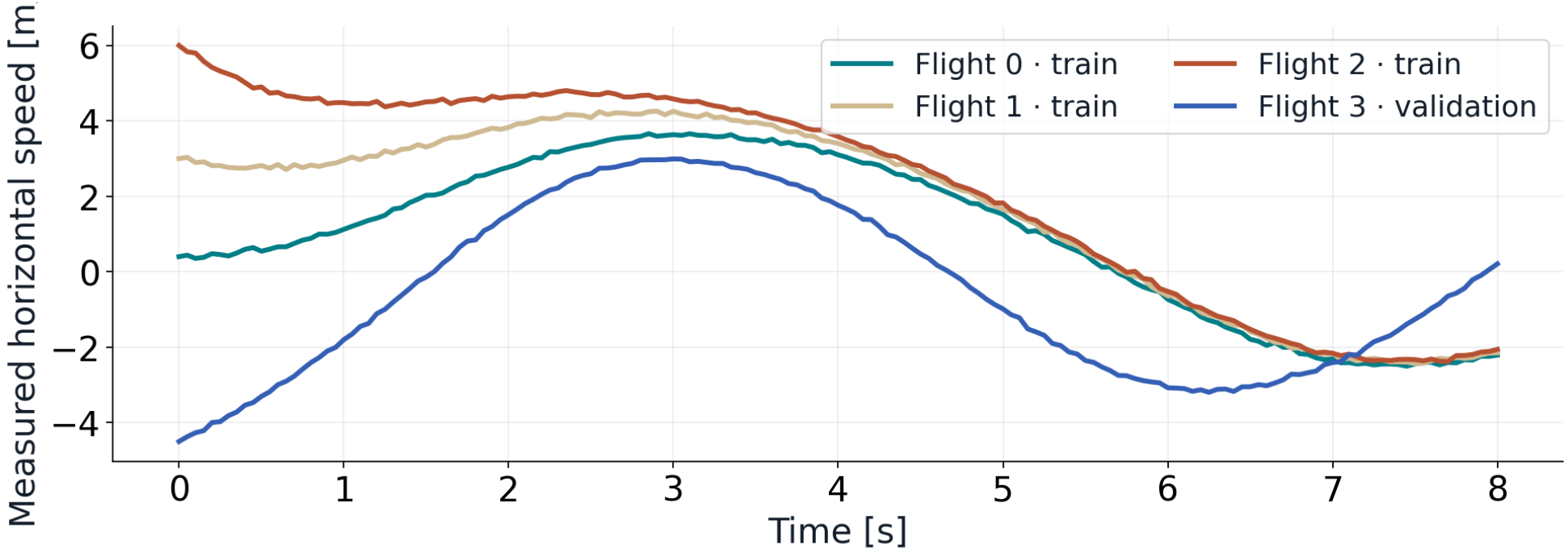
Your simulation plot is the recovery point.

Next: learn drag from noisy data.

Worksheet and commands: WORKSHOP.md · scan the opening QR.

Known experiment. Unknown coefficients.

Three training flights; one held-out flight. 20 Hz, 8 s each.



Speed variation helps separate linear and quadratic drag.

Differentiate the prediction error

Positive parameters; noise-scaled loss; bounded optimization

Generated RHS Modelica → JAX → RK4 rollout positive drag → Data residuals scaled by noise → JAX gradients L-BFGS-B

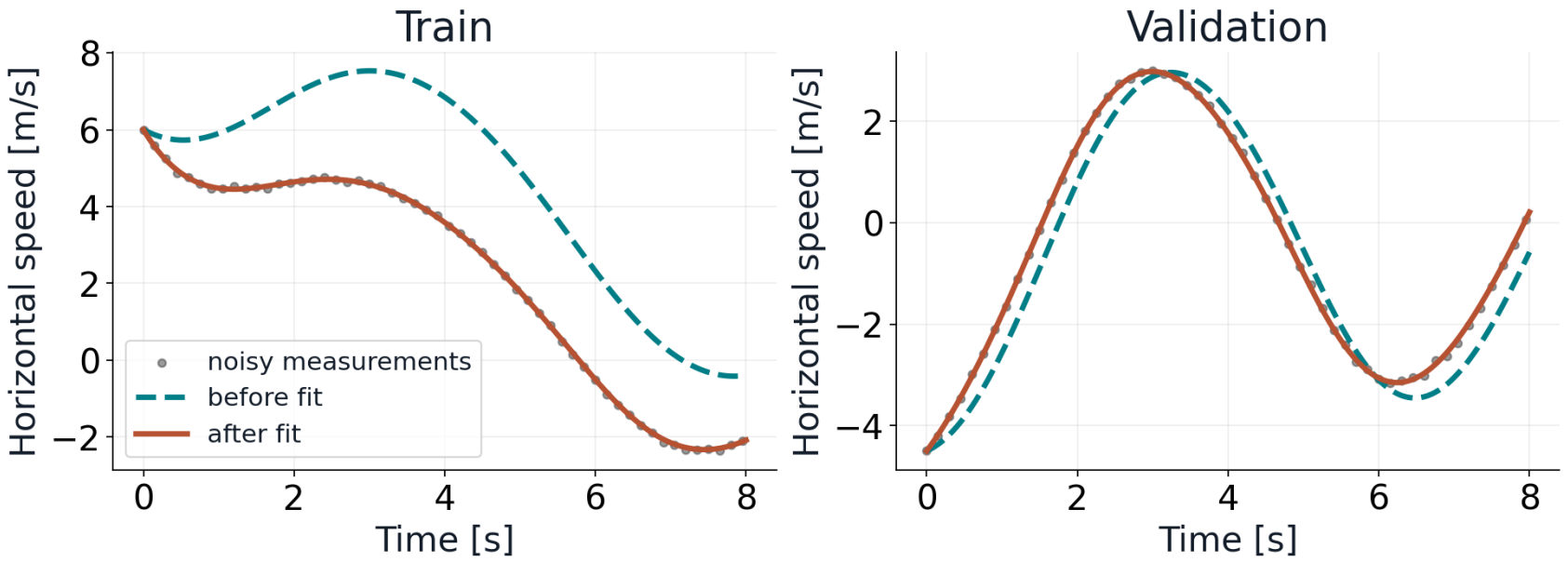
$$L = \mathrm{mean}\left[\left(\frac{\hat{y} - y}{\sigma}\right)^2\right]$$

$$k_l / k_{l,\mathrm{ref}} = e^a, \quad k_q / k_{q,\mathrm{ref}} = e^b$$

JAX provides gradients. SciPy updates the coefficients.

Run the fit. Test the held-out flight.

Open fit_drag.png and fit_drag.json; compare before and after.




```
$ uv run --frozen python examples/fit_drag_jax.py
```

Two coefficients · lower loss · improved held-out prediction

Check your result. Keep building.

01 Open your before-and-after prediction plot

02 Compare the held-out flight and fitted values

03 Stretch: rerun from another starting guess

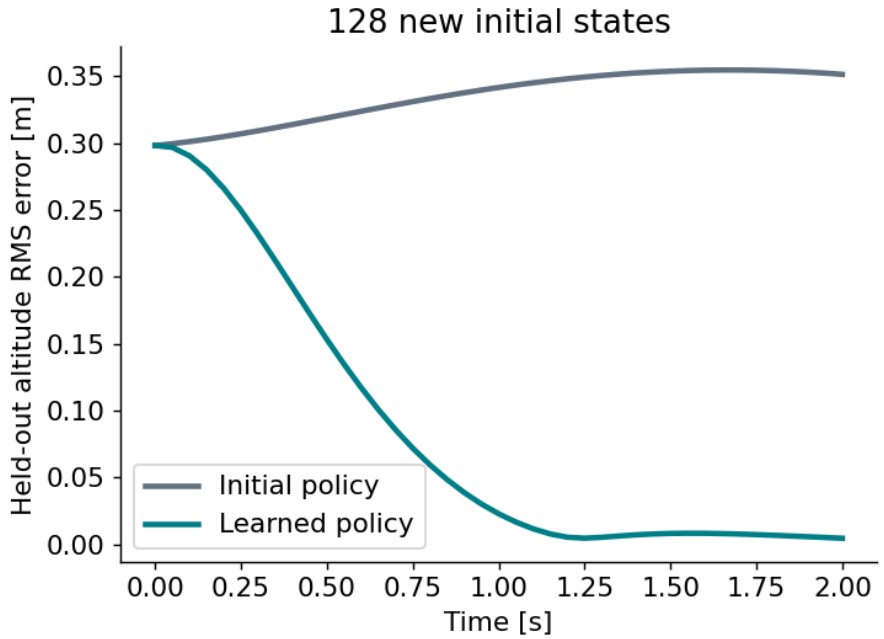
04 App builders: keep your prototype moving

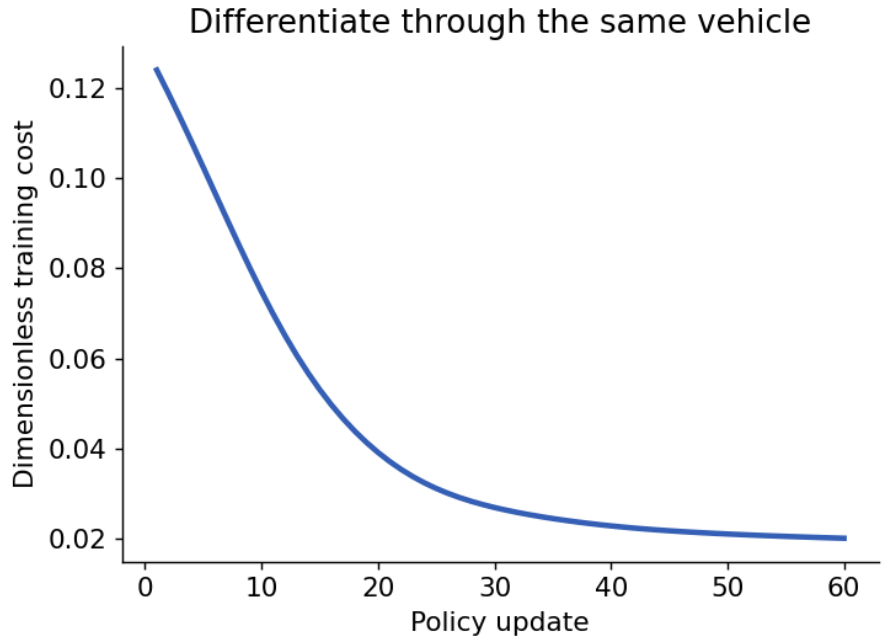
Fit complete? Continue your experiment or explore the optional checks.

Workshop guide ↗

Make a target. Learn a policy.

target.toml → generated Torch → differentiable policy learning






```
$ uv run --frozen --group torch python scripts/learn_hover_torch.py
```

Held-out cost ≈ 0.116 → 0.021 · optional instructions: BACKENDS.md

Backend walkthrough ↗ | Torch target.toml ↗ | Lean Jinja target ↗

Back at

12:10

10 MINUTES

Stretch. Ask a question. Keep your editor open.

KEEP YOUR PLACE

Keep your fitted coefficients.

Next: plan a flight using your fit.

Your fit plot is the recovery point.

Same Modelica source; now compile to CasADi.

Move between hovers, respect limits

THE IDEA

1. (0, 1) m → (4, 2) m in 6 s
2. Track target; limit effort and control changes
3. Bound thrust, torque, pitch and altitude



T: 0–20 N τ: ±0.35 N m

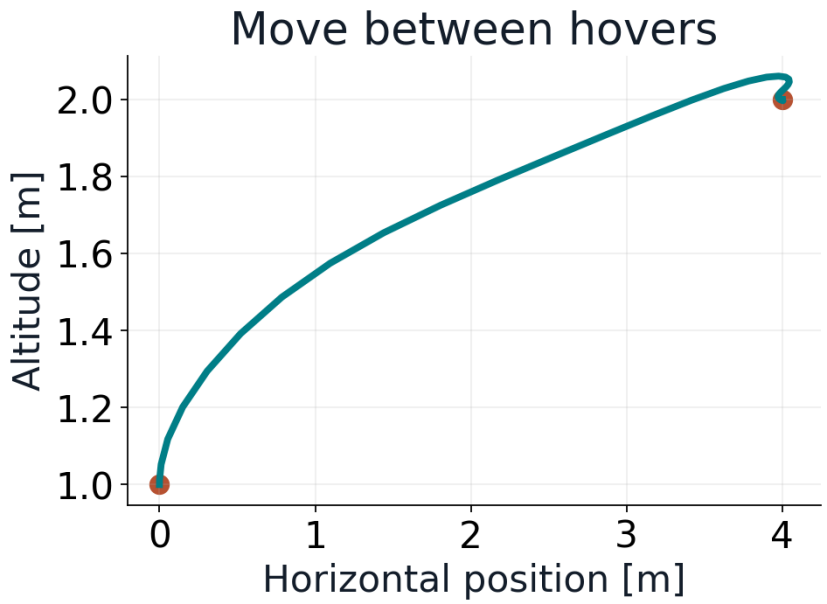
z: 0.25–4 m |θ| ≤ 0.55 rad

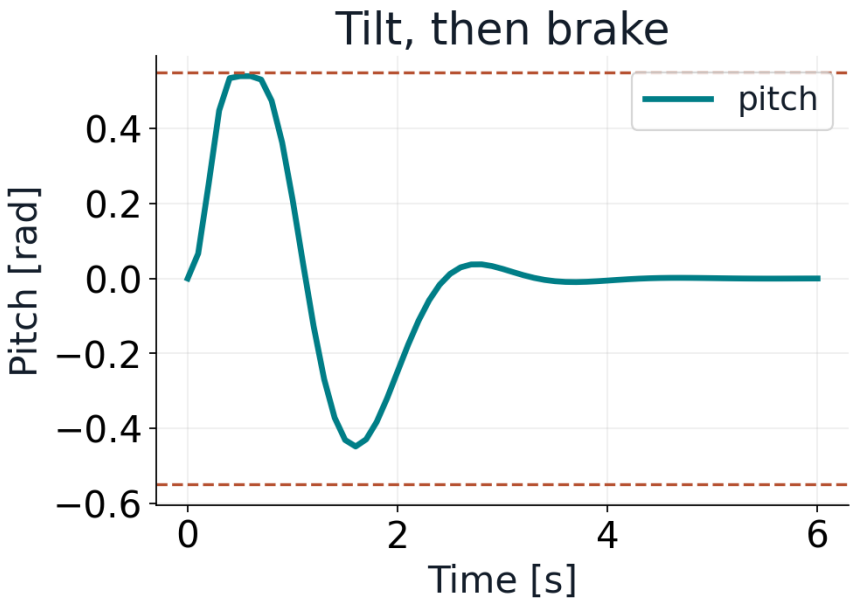
60 shooting intervals · node pitch margin · independent reintegration

Identified drag is an input to this optimization.

Solve. Inspect. Explain the braking.

Open optimization.png and optimization.json.






```
$ uv run --frozen python examples/optimize_casadi.py
```

Solve_Succeeded · limits respected · arrive near rest

Workshop guide ↗

Next up: pylon racing at PURT

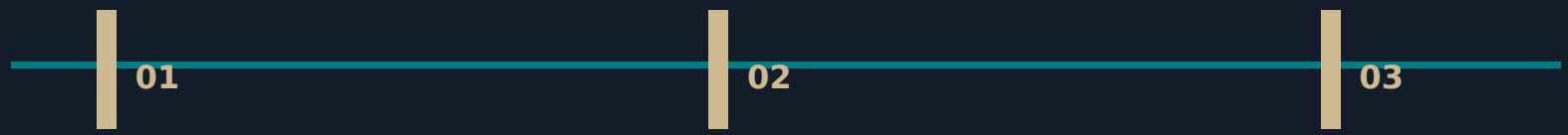
FROM DIGITAL TWIN TO THE RACE COURSE

Build. Fly. Race.

The next competition will leverage Rumoca

Design, simulate, then flight-test your autonomy

Digital twin → real aircraft at PURT



ILLUSTRATIVE COURSE · EVENT DETAILS ON THE SITE

FOLLOW THE
NEXT COMPETITION

pylon-racing.geddes.rcac.purdue.edu

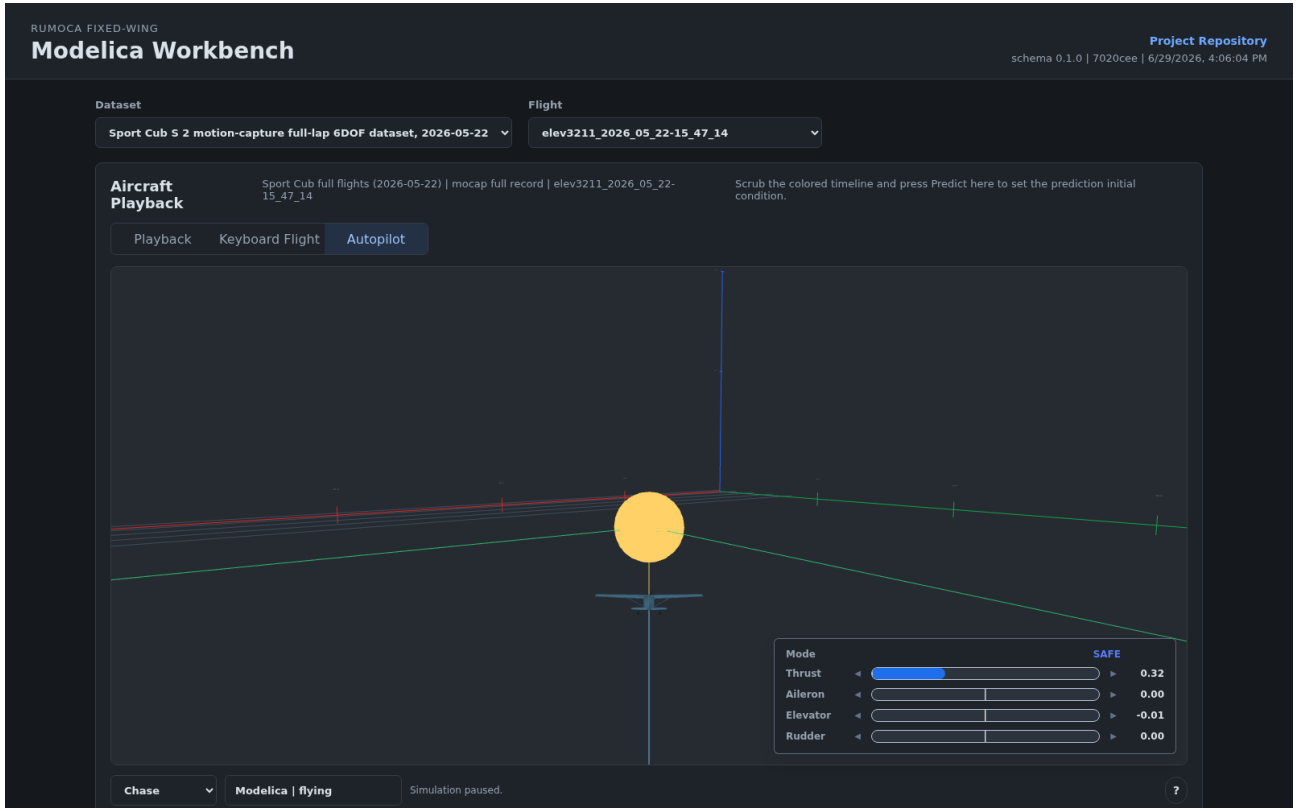
PURT pylon racing ↗

Challenge rules ↗

Explore the fixed-wing workbench

TRY THIS

1. Modelica flight simulation + autopilot
2. Try 50 Hz → 20 Hz, restart, then observe
3. Also: slam_web (in development), mocap_sysid



Live links + short experiments: BROWSER_APPS.md

Fixed-wing demo ↗ | Fixed-wing code ↗ | Mocap demo ↗ | Mocap code ↗ | SLAM (in dev) ↗ | SLAM code ↗

Show your experiment. Vote for the cap.

5	45 s	1
MAX. PRESENTERS	EACH DEMO	VOTE PER PERSON

Show your change + one checked result

Name your tool. Share elapsed work time.

Audience favorite wins a free CogniPilot stocking cap

Random draw if more than 5 ready entries → audience hand vote

Challenge rules ↗ Optional gallery ↗

Bring your model.
Join the work.

1. Bring a model or a question from your own domain
2. Add a validation case, model library or compiler fix
3. Start a conversation: problem + model + one check

James Goppert · Micah Condie

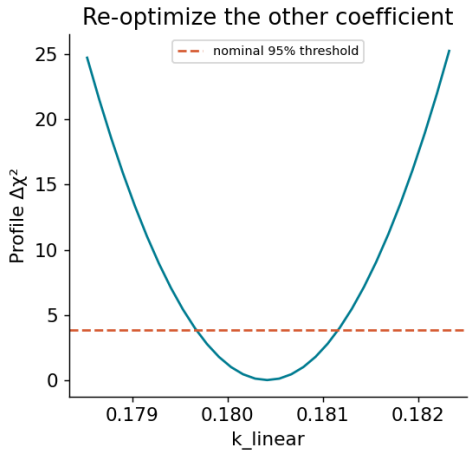
Scan to join the work · COMMUNITY.md · questions welcome

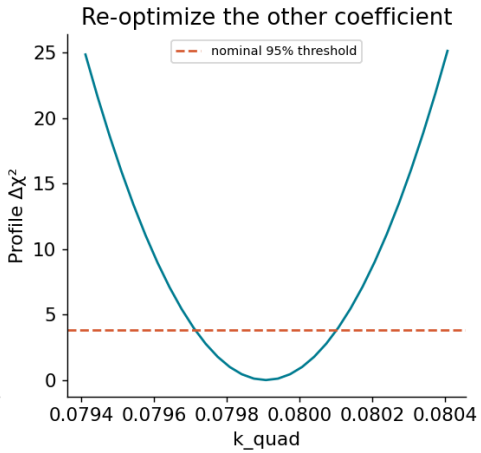
Join the community ↗ **Rumoca compiler ↗** **Model library ↗**

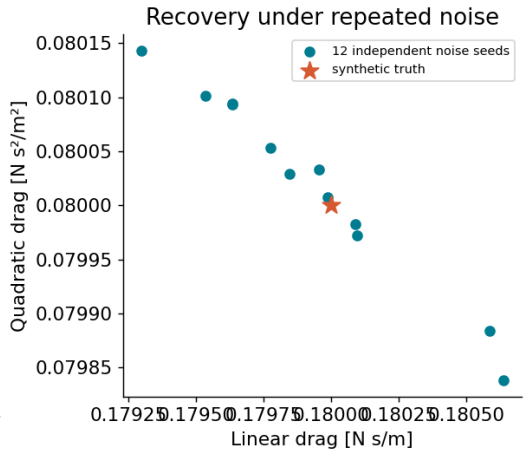
Exact-real proofs and numerical evidence

THE IDEA

1. Lean: equation equality and invariants
2. Audit: gradients, profiles, integration
3. Papers: model and method choices







None of these certify arbitrary edits or real flight.

Lean proof source ↗ | Math evidence ↗

Primary papers behind the approach

1. Faessler et al., RA-L 2018: rotor-drag physics and excitation
2. Andersson et al., 2019: CasADi AD and multiple shooting
3. Raue 2009; Wieland 2021: practical identifiability and profiles

World-axis teaching drag differs from body-axis rotor aerodynamics.

FastDyn: fly the firmware against declared physics

01 Rumoca compiles the Modelica plant into an FMI 3 FMU.

02 FastDyn couples sensors and actuators to ArduPilot firmware in QEMU.

03 Instruction-counted virtual time coordinates firmware and plant steps.

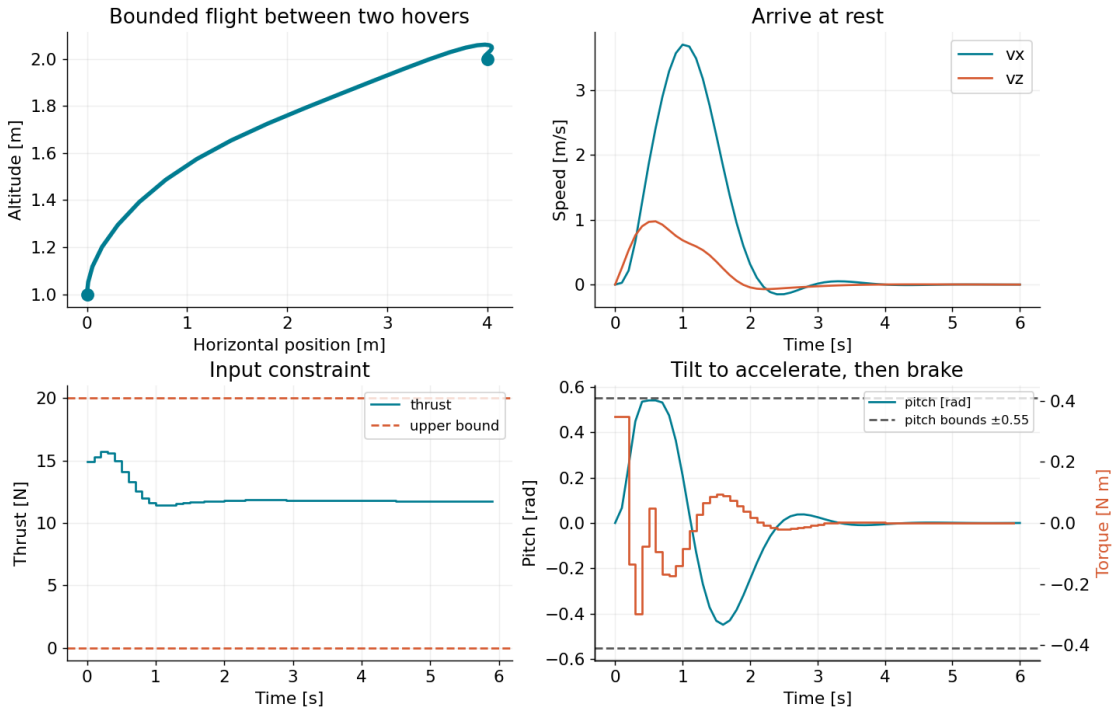
04 Optional: three load trials, isolated logs and trajectory comparisons.

Separate FastDyn Codespace · keep the core setup lightweight.

FastDyn demo guide ↗ | BoardRunner paper ↗ | FastDyn source ↗

Numerical constraints need validation

Reintegrate the inputs independently; inspect errors by physical unit.



Small defects, pitch margin and fine rollout are separate checks.

Numerical evidence ↗